

Vector Operations in MATLAB

Student handout - code first, followed by detailed explanation

1. MATLAB Code

Copy the following script into a MATLAB file, for example **vectorOperations_student_version.m**, and run it from the Editor.

```
01 % Purpose: Demonstrate how vectors interact mathematically in MATLAB
02
03 clear
04 close all
05 clc
06
07 %% Part 1 - Basic vector operations
08
09 % Define two row vectors and one column vector
10 rvec1 = [6, 0, 2, 5];
11 rvec2 = [9, 9, 1, 9];
12 cvec = [9; 4; 8; 1];
13
14 fprintf('Addition of two row vectors:\n')
15 disp(rvec1 + rvec2)
16
17 fprintf('Subtracting a scalar from a row vector:\n')
18 disp(rvec1 - 3)
19
20 fprintf('Element-wise division of two row vectors:\n')
21 disp(rvec1 ./ rvec2)
22
23 fprintf('Addition of a row vector and a column vector:\n')
24 disp(rvec1 + cvec)
25
26 %% Part 2 - Matrix multiplication with row and column vectors
27
28 % Define two column vectors
29 cvec1 = [9; 4; 8; 1];
30 cvec2 = [4; 3; 8; 6];
31
32 fprintf('Column vector times row vector (outer product):\n')
33 disp(cvec1 * rvec1)
34
35 fprintf('Row vector times column vector (dot product):\n')
36 disp(rvec1 * cvec1)
37
38 % The following operation is not valid because both vectors are 4 x 1:
39 % disp(cvec1 * cvec2)
40
41 fprintf('Correct dot product of two column vectors using transpose:\n')
42 disp(cvec1' * cvec2)
43
44 fprintf('Element-wise multiplication of two column vectors:\n')
45 disp(cvec1 .* cvec2)
```

2. Detailed Explanation

Learning objective: This script shows how MATLAB treats row vectors and column vectors in arithmetic operations, element-wise operations, and matrix multiplication.

2.1 Clearing the previous MATLAB session

The commands `clear`, `close all`, and `clc` prepare MATLAB for a clean run. `clear` removes old variables from the Workspace, `close all` closes open figure windows, and `clc` clears the Command Window.

2.2 Row vectors and column vectors

Variable	MATLAB definition	Size	Description
<code>rvec1</code>	<code>[6, 0, 2, 5]</code>	1 x 4	A row vector with four elements.
<code>rvec2</code>	<code>[9, 9, 1, 9]</code>	1 x 4	Another row vector with the same size as <code>rvec1</code> .
<code>cvec</code>	<code>[9; 4; 8; 1]</code>	4 x 1	A column vector. Semicolons create new rows.

In MATLAB, commas or spaces separate elements in the same row. Semicolons separate rows. Therefore, `[6, 0, 2, 5]` is a row vector, while `[9; 4; 8; 1]` is a column vector.

2.3 Addition of two row vectors

The command `rvec1 + rvec2` adds corresponding elements because both vectors have the same size, 1 x 4.

`[6, 0, 2, 5] + [9, 9, 1, 9] = [15, 9, 3, 14]`

2.4 Subtracting a scalar from a vector

The command `rvec1 - 3` subtracts 3 from every element of `rvec1`. MATLAB automatically applies the scalar to all elements of the vector.

`[6, 0, 2, 5] - 3 = [3, -3, -1, 2]`

2.5 Element-wise division

The command `rvec1 ./ rvec2` uses the dot operator. This means MATLAB divides element by element, not as matrix division.

`[6/9, 0/9, 2/1, 5/9] = [0.6667, 0, 2.0000, 0.5556]`

Important rule: in MATLAB, `.*`, `./`, and `.^` are used for element-wise multiplication, division, and powers.

2.6 Adding a row vector and a column vector

The command `rvec1 + cvec` adds a 1 x 4 row vector to a 4 x 1 column vector. In recent MATLAB versions, this uses implicit expansion and produces a 4 x 4 matrix. Each element of the column vector is added to the entire row vector.

`rvec1 + cvec =`

```
[6 0 2 5] + 9 -> [15 9 11 14]
[6 0 2 5] + 4 -> [10 4 6 9]
[6 0 2 5] + 8 -> [14 8 10 13]
[6 0 2 5] + 1 -> [ 7 1 3 6]
```

2.7 Matrix multiplication: column times row

The command `cvec1 * rvec1` multiplies a 4 x 1 column vector by a 1 x 4 row vector. The inner dimensions match, so the operation is valid. The result has size 4 x 4. This is called an outer product.

$$(4 \times 1) * (1 \times 4) = 4 \times 4$$

2.8 Matrix multiplication: row times column

The command `rvec1 * cvec1` multiplies a 1×4 row vector by a 4×1 column vector. The result has size 1×1 , so it is a single number. This is the dot product.

$$6(9) + 0(4) + 2(8) + 5(1) = 75$$

2.9 Why column times column is not valid with *

The original operation `cvec1 * cvec2` tries to multiply a 4×1 vector by another 4×1 vector. This is not valid matrix multiplication because the inner dimensions do not match.

$(4 \times 1) * (4 \times 1)$ is invalid because 1 is not equal to 4.

To calculate the dot product of two column vectors, transpose the first vector:

$$\text{cvec1}' * \text{cvec2} = 118$$

To multiply the two column vectors element by element, use the dot operator:

$$\text{cvec1} .* \text{cvec2} = [36; 12; 64; 6]$$

3. Expected Command Window Output

Students can compare their MATLAB output with the following results. Minor formatting differences are normal depending on MATLAB display settings.

Addition of two row vectors:

```
15     9     3    14
```

Subtracting a scalar from a row vector:

```
3    -3    -1     2
```

Element-wise division of two row vectors:

```
0.6667     0    2.0000    0.5556
```

Addition of a row vector and a column vector:

```
15     9    11    14
10     4     6     9
14     8    10    13
7      1     3     6
```

Column vector times row vector (outer product):

```
54     0    18    45
24     0     8    20
48     0    16    40
6      0     2     5
```

Row vector times column vector (dot product):

```
75
```

Correct dot product of two column vectors using transpose:

```
118
```

Element-wise multiplication of two column vectors:

```
36
12
64
6
```

4. Key Rules to Remember

Rule	Example / Note
Same-size vectors can be added or subtracted element by element.	Example: <code>rvec1 + rvec2</code> .
A scalar can operate on every element of a vector.	Example: <code>rvec1 - 3</code> .
Use dot operators for element-wise operations.	Examples: <code>.*</code> , <code>./</code> , and <code>.^</code> .
For matrix multiplication with <code>*</code> , the inner dimensions must match.	Example: $(1 \times 4) * (4 \times 1)$ is valid.
A row vector times a column vector gives a scalar dot product.	Example: <code>rvec1 * cvec1 = 75</code> .
A column vector times a row vector gives a matrix outer product.	Example: <code>cvec1 * rvec1</code> gives a 4×4 matrix.
Two column vectors cannot be multiplied directly using <code>*</code> unless one is transposed.	Correct example: <code>cvec1' * cvec2</code> .

Instructor note: The line `disp(cvec1*cvec2)` in the original script was intentionally corrected here because it produces a MATLAB dimension error. The handout keeps the concept but shows the correct alternatives for dot product and element-wise multiplication.